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INFERENTIAL STATISTICS PROJECT

BUSSINESS REPORT

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1. Introduction

In an era where data drives decisions, statistical analysis serves as a critical tool to uncover patterns, test assumptions, and guide improvements across industries. This report tackles four practical challenges—spanning sports, manufacturing, geology, and healthcare—through a structured, analytical lens. By applying hypothesis testing, regression analysis, and data visualization techniques, we aim to extract meaningful insights and deliver actionable recommendations for:

1. **Football Injury Patterns**: Analysing injury trends to identify risk factors and propose strategies for player safety.
2. **Gunny Bag Durability**: Assessing material strength to ensure reliability in packaging and logistics.
3. **Stone Hardness for Printing**: Evaluating geological suitability for high-precision industrial applications.
4. **Dental Implant Performance**: Investigating material properties to enhance durability and patient outcomes.

Each analysis is grounded in rigorous statistical methodologies, supported by visualizations to clarify trends and relationships. While the findings are rooted in the provided datasets, we acknowledge the inherent limitations of scope and methodology, ensuring transparency in our conclusions.

This report bridges data and real-world impact, offering stakeholders evidence-based strategies to refine processes, reduce inefficiencies, and innovate solutions. Let's explore how numbers can inform smarter decisions—and better outcomes.

2. A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

The dataset has 4 rows and 2 columns. It is always a good practice to view a sample of the rows. A simple way to do that is to use head() function

	Position	Players Injured	Players Not Injured
0	Striker	45	32
1	Forward	56	38
2	Attacking Midfielder	24	11
3	Winger	20	9

- **Descriptive Statistics**

	Players Injured	Players Not Injured
count	4.000000	4.000000
mean	36.250000	22.500000
std	17.134274	14.662878
min	20.000000	9.000000
25%	23.000000	10.500000
50%	34.500000	21.500000
75%	47.750000	33.500000
max	56.000000	38.000000

Observations:

The data reveals that the average number of injured players (36.25) is notably higher compared to the average number of non-injured players (22.5), highlighting a concerning trend of frequent injuries within the team.
The wider range and higher standard deviation among injured players suggest that injuries are not evenly spread, with certain positions being more prone to injuries than others. On the other hand, while the distribution of non-injured players is relatively more consistent, it still exhibits considerable variability, indicating room for further analysis into factors influencing player availability.
These statistics highlight the importance of understanding positional factors contributing to injuries, which could inform targeted interventions by the physiotherapist.

2.1 What is the probability that a randomly chosen player would suffer an injury?

Analysis:

Total number of players injured: 145

Total number of players: 235

Probability = Total injured players / Total players

Calculation: 145/235 = 0.617

The probability that a randomly chosen player would suffer an injury is 0.617 or 61.7%

2.2 What is the probability that a player is a forward or a winger?

Analysis:

Number of forwards: 94 Number of wingers: 29

Total players in these positions: 123

Probability = Players in forward or winger positions / Total players

Calculation: 123/235 = 0.523

The probability that a player is a forward or a winger is: 0.523 or 52.3%

2.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

Analysis:

Number of injured strikers: 45

Total number of players: 235

Probability = Injured strikers / Total players

Calculation: 45/235 = 0.191

The probability that a randomly chosen player plays in a striker position and has a foot injury: 0.191 or 19.1%

2.4 What is the probability that a randomly chosen injured player is a striker?

Analysis: Number of injured strikers: 45

Total number of injured players: 145

Probability = Injured strikers / Total injured players

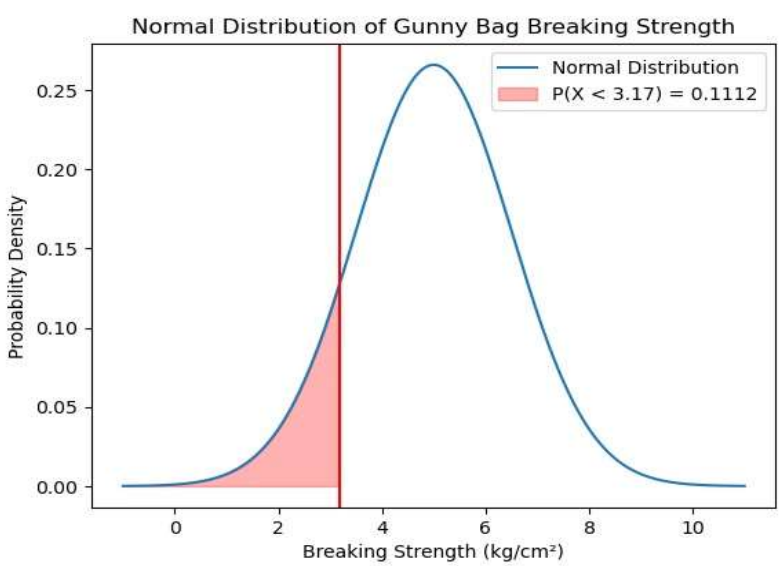
Calculation: 45/145 = 0.31

Probability that a randomly chosen injured player is a striker: 0.310 or 31%

3. The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information;

3.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq. cm?

To determine the proportion of gunny bags with a breaking strength less than 3.17 kg/cm², we utilized the cumulative distribution function (CDF) of the normal distribution. The CDF provides the probability that a random variable (in this case, breaking strength) is less than or equal to a specified value.



Where:
P (X ≤ x) is the proportion of gunny bags with a breaking strength less than or equal to x kg/cm².

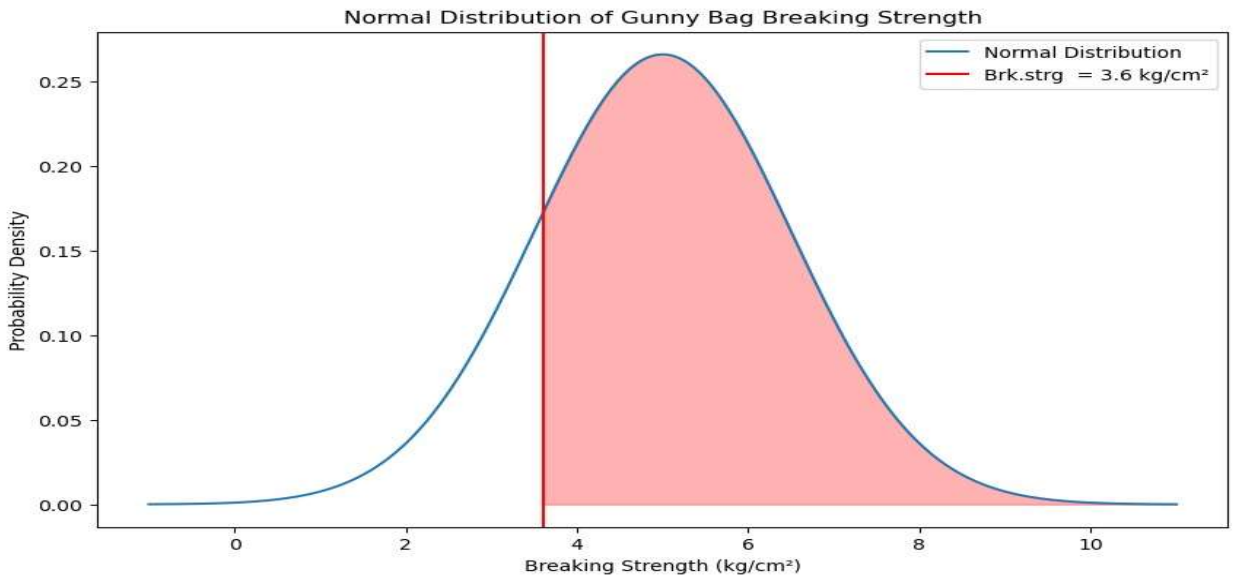
Specified breaking strength (x): 3.17 kg/cm²

After computation, the proportion was found to be approximately (0.1112)

This calculation indicates that about 11.2 percent of gunny bags in our dataset have a breaking strength below 3.17 kg/cm².

3.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq. cm?

To determine the proportion of gunny bags with a breaking strength of at least 3.6 kg/cm², we employed the complementary cumulative distribution function (CDF) of the normal distribution. The CDF provides the probability that a random variable (breaking strength) is greater than a specified value.



Where:

P (X ≥ x) is the proportion of gunny bags with a breaking strength greater than or equal to x kg/cm².

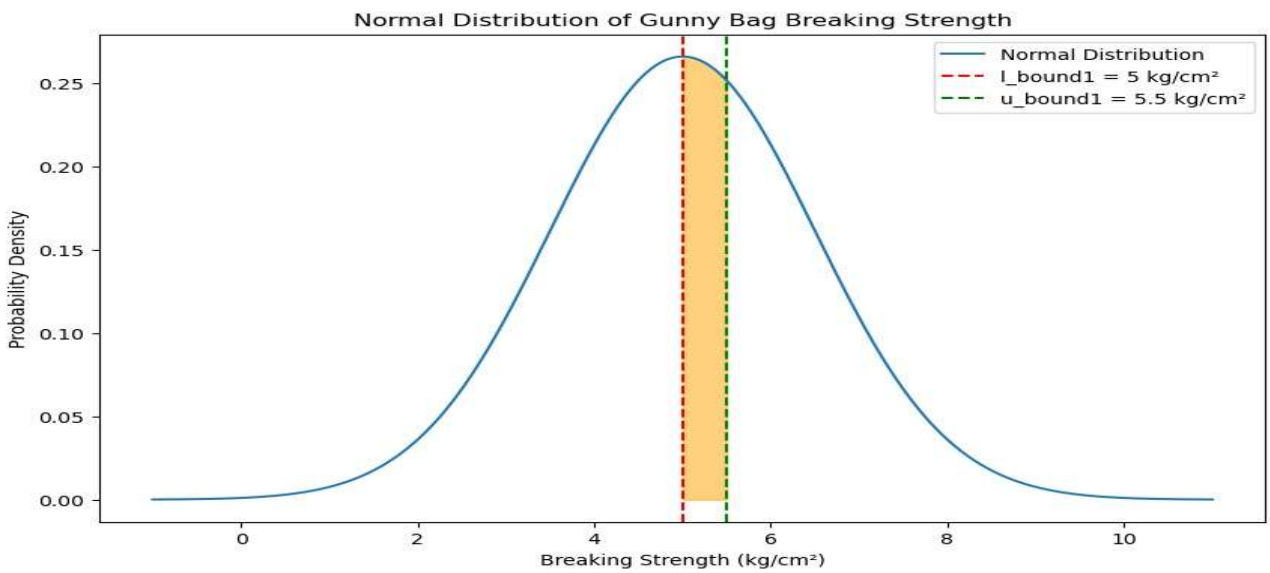
Specified breaking strength (x): 3.6 kg/cm²

After computation, the proportion was found to be approximately (0.8247)

This calculation indicates that approximately 82.5 percent of gunny bags in our dataset have a breaking strength of at least 3.6 kg/cm².

3.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq. cm?

To determine the proportion of gunny bags with a breaking strength between 5 and 5.5 kg/cm², we utilized the cumulative distribution function (CDF) of the normal distribution. The CDF provides the probability that a random variable (in this case, breaking strength) falls below a specified value.



Where:

P (l ≤ X ≤u) is the proportion of gunny bags with a breaking strength between l and u kg/cm².

Lower bound (l): 5 kg/cm²

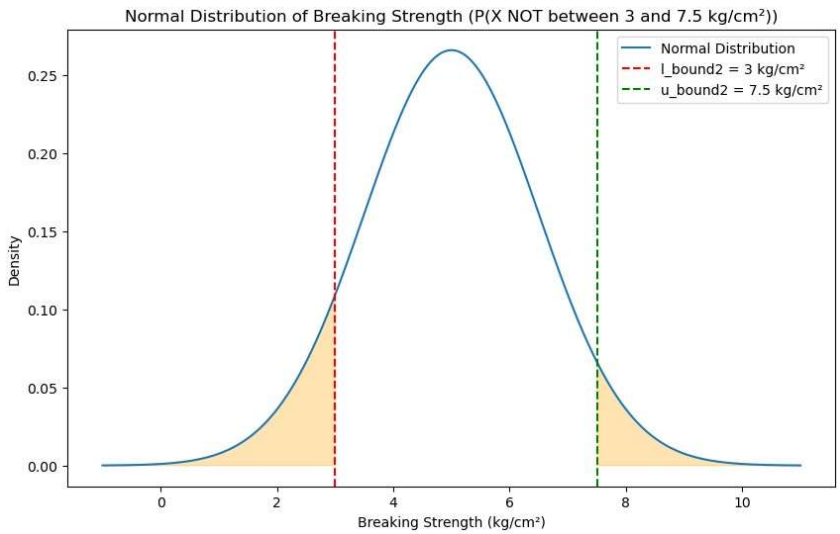
Upper bound (u): 5.5 kg/cm²

After computation, the proportion was found to be approximately (0.1306)

This calculation indicates that approximately 13 percent of gunny bags in our dataset have a breaking strength between 5 and 5.5 kg/cm².

3.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq. cm?

To determine the proportion of gunny bags with a breaking strength not between 3 and 7.5 kg/cm², we used the cumulative distribution function (CDF) of the normal distribution. The CDF provides the probability that a random variable (in this case, breaking strength) falls below a specified value.



Where:

P (X < l or X > u) is the proportion of gunny bags with a breaking strength not between l and u kg/cm².

Mean breaking strength (μ): 5 kg/cm²

Standard deviation (σ): 1.5 kg/cm²

Lower bound (l): 3 kg/cm²

Upper bound (u): 7.5 kg/cm²

After computation, the proportion was found to be approximately (0.1390)

This calculation indicates that approximately 13.9 percent of gunny bags in our dataset have a breaking strength not between 3 and 7.5 kg/cm².

4. Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

Data Description

Dataset contains data with a batch of polished and unpolished stones

Data Overview

The dataset has 75 rows and 2 columns.

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 75 entries, 0 to 74
Data columns (total 2 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Unpolished           75 non-null     float64
 1   Treated and Polished 75 non-null     float64
dtypes: float64(2)
memory usage: 1.3 KB
```

A quick look at the dataset information tells us that there are 2 numerical and 0 categorical variables. There are no Null records present in the database.

Statistical Summary

Inspecting the Summary Statistics of the Dataset (Numerical fields)

	Unpolished	Treated and Polished
count	75.000000	75.000000
mean	134.110527	147.788117
std	33.041804	15.587355
min	48.406838	107.524167
25%	115.329753	138.268300
50%	135.597121	145.721322
75%	158.215098	157.373318
max	200.161313	192.272856

Observations:

1. Increased Hardness in Treated and Polished Stones:
   Treated and polished stones show higher mean and median hardness values than unpolished stones, indicating that the treatment and polishing processes significantly improve stone hardness. This enhancement makes them more suitable for applications requiring durability.
2. Reduced Variability in Treated and Polished Stones:
   The standard deviation for treated and polished stones (15.59) is much lower than that of unpolished stones (33.04). This suggests that the treatment process yields stones with more consistent hardness levels, which is ideal for applications needing uniformity.
3. Greater Consistency in Hardness:
   The interquartile range (IQR) for treated and polished stones (19.10) is narrower compared to unpolished stones (42.89). This further emphasizes the consistency in hardness achieved through treatment and polishing, ensuring reliable performance.
4. Broader Hardness Range in Unpolished Stones:
   Unpolished stones exhibit a wider range of hardness values (48.41 to 200.16), with some stones having significantly lower hardness. This variability indicates that unpolished stones may not meet the standards required for high-hardness applications, making them less dependable.
5. Improved Quality Through Treatment:
   The mean hardness of stones rises from 134.11 (unpolished) to 147.79 (treated and polished), showcasing the effectiveness of the treatment and polishing process in enhancing both the quality and performance of the stones.

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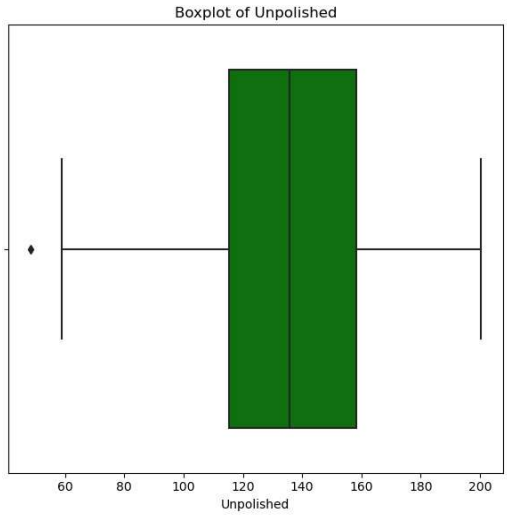
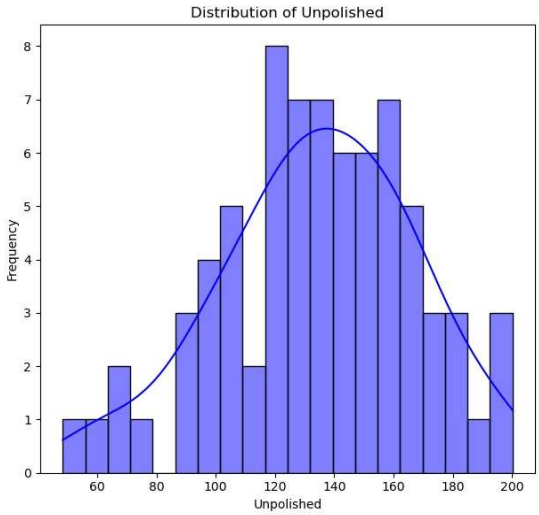
Univariate Analysis

For performing Univariate analysis, we will take a look at the Boxplots and Histograms to get a better understanding of the distributions.

Numerical variables

- **Observations on Unpolished**

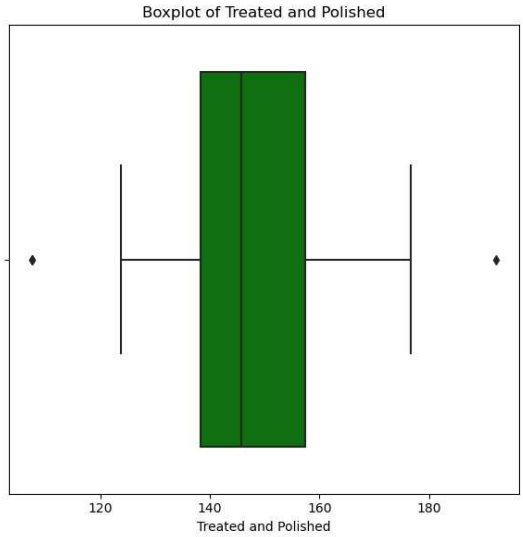


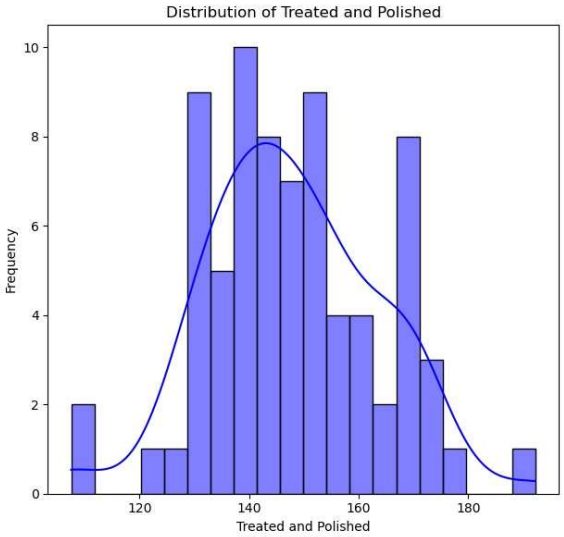


Observations:

- The Distribution of **Unpolished** is slightly left skewed.
- There is an outlier in this variable, we will treat the outlier.

Observations on Treated and Polished Stones





Observations:

- The Distribution of **Treated and Polished Stones** is slightly left skewed.
- There are a few outliers in this variable, we will treat the outliers.

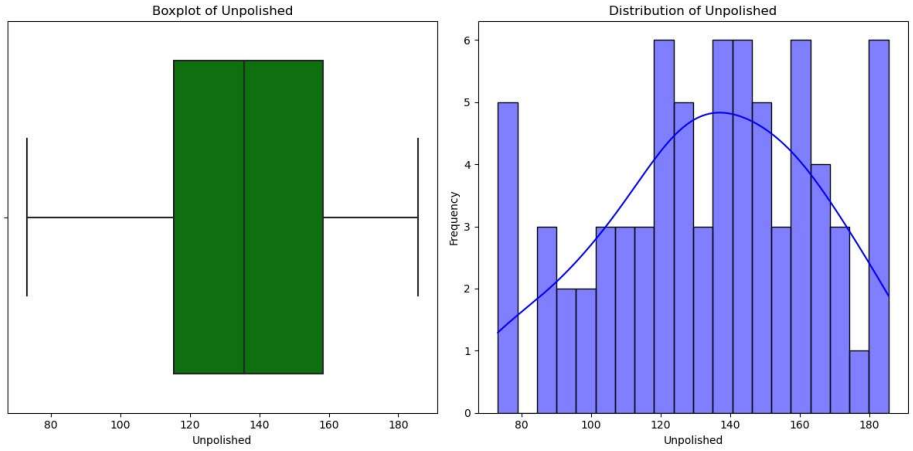
Referring to the above box plots it seems that the two variables Unpolished and Treated and Polished have outliers present in the variables.

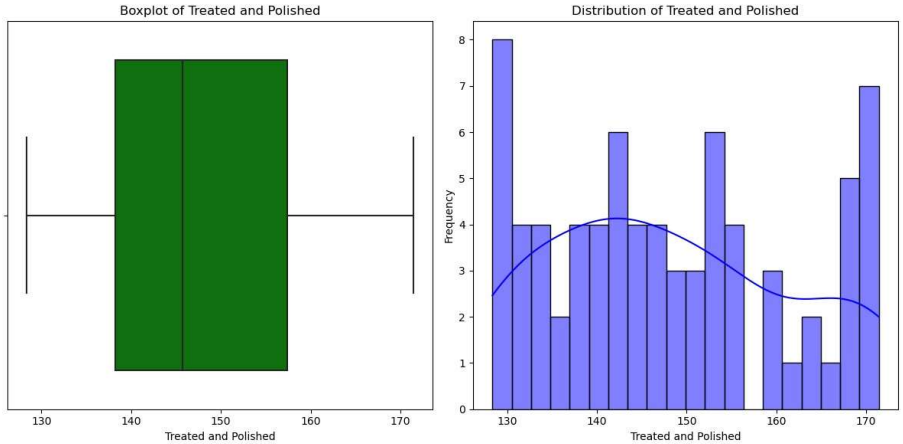
These outlier values need to be treated and there are several ways of treating them: **

- Drop the outlier value
- Replace the outlier value using the IQR
- We can treat the outlier's using percentile or with Inter quartile Range

Here, we are using percentile to replace outliers

Boxplots and Histograms after outliers' treatment





4.1 Suitability of Unpolished Stones for Printing

- **Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?**

	Unpolished	Treated and Polished
count	75.000000	75.000000
mean	134.334853	148.064394
std	30.623973	13.459406
min	73.178331	128.357069
25%	115.329753	138.268300
50%	135.597121	145.721322
75%	158.215098	157.373318
max	185.582012	171.416767

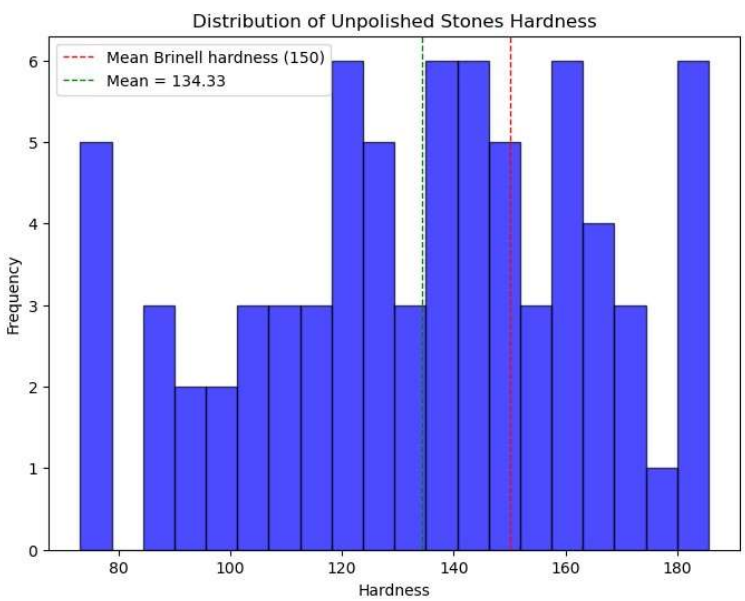
To determine whether Zingaro is justified in thinking that the unpolished stones may not be suitable for printing, we can perform a hypothesis test on the mean Brinell hardness index of the unpolished stones. We'll assume a significance level of 5% ($\alpha = 0.05$). The null hypothesis (H_0) and the alternative hypothesis (H_1) can be stated as follows:

Null Hypothesis (H 0): The mean Brinell hardness of unpolished stones is at least 150. ($\mu \geq 150$)

Alternative Hypothesis (H 1): The mean Brinell hardness of unpolished stones is less than 150. ($\mu < 150$)

Sample Data:

- Sample Mean (x): 134.33
- Sample Standard Deviation (s): 30.62
- Sample Size (n): 75
- Population Mean (μ): 150



The t-statistic for a one-sample t-test is calculated as $t \approx -4.43$

The one-tailed p-value is approximately 0.00002

Decision Rule:

At the 5% significance level (α=0.05), if the p-value is less than 0.05 and the t-statistic is negative, we reject the null hypothesis.

Since the one-tailed p-value is approximately 0.00002, which is much less than 0.05, and the t-statistic is -4.43, which is negative, we reject the null hypothesis. This provides sufficient evidence to conclude that the mean Brinell hardness index of unpolished stones is significantly less than 150.

There is strong evidence to support the claim that the unpolished stones have a mean Brinell hardness index less than 150. Therefore, Zingaro is justified in thinking that the unpolished stones may not be suitable for printing.

4.2 Comparison of Mean Hardness Between Polished and Unpolished Stones

Is the mean hardness of the polished and unpolished stones the same?

Null Hypothesis(H0): The mean hardness of polished stones is equal to the mean hardness of unpolished stones

Alternative Hypothesis(H1): The mean hardness of polished stones is not equal to the mean hardness of unpolished stones**.**

Polished Stones:

Sample size: 75

Mean hardness: 148.06

Standard deviation: 13.46

Unpolished Stones:

Sample size: 75

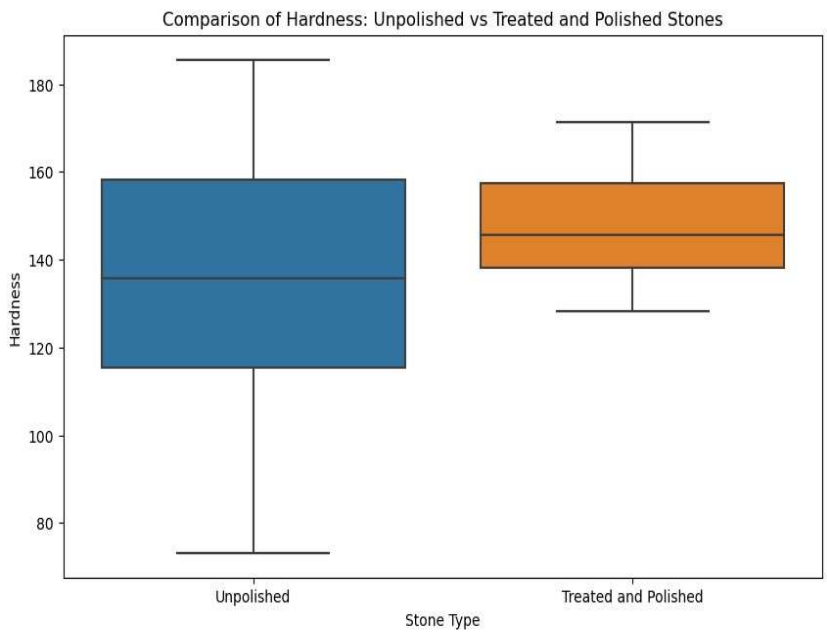
Mean hardness: 134.33

Standard deviation: 30.62

Test Statistic (t-value): $t \approx 3.55$

Degrees of Freedom (df): 148

Critical t-value (at α=0.05): ±1.976



Results:

The calculated t-value (3.49) exceeds the critical t-value (±1.976), leading to the rejection of the null hypothesis. Therefore, there is a statistically significant difference in the mean hardness of polished and unpolished stones.

Conclusion: The findings suggest that the level of hardness varies between polished and unpolished stones, which could have implications for the printing process at Zingaro Stone Printing. Further investigation into the factors affecting stone hardness may be warranted to ensure optimal printing outcomes.

5. Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest.

Data Description

Dental_Hardness_data.csv dataset contains Dental implant data.

Data Overview

Load the required packages, set the working directory, and load the data file.

The dataset has 107 rows and 14 columns.

	Dentist	Method	Alloy	Temp	Response	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10	Unnamed: 11	Unnamed: 12	Unnamed: 13
0	1.0	1.0	1.0	1500.0	813.0	NaN	NaN	Anova: Two-Factor Without Replication	NaN	NaN	NaN	NaN	NaN	NaN
1	1.0	1.0	1.0	1600.0	792.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	1.0	1.0	1.0	1700.0	792.0	NaN	NaN	SUMMARY	Count	Sum	Average	Variance	NaN	NaN
3	1.0	1.0	2.0	1500.0	907.0	NaN	NaN	1	4	2315	578.75	523721.583333	NaN	NaN
4	1.0	1.0	2.0	1600.0	792.0	NaN	NaN	1	4	2394	598.5	584819	NaN	NaN

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 107 entries, 0 to 106
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Dentist       90 non-null     float64
 1   Method        90 non-null     float64
 2   Alloy         90 non-null     float64
 3   Temp          90 non-null     float64
 4   Response      90 non-null     float64
 5   Unnamed: 5    0 non-null      float64
 6   Unnamed: 6    0 non-null      float64
 7   Unnamed: 7    102 non-null    object
 8   Unnamed: 8    100 non-null    object
 9   Unnamed: 9    100 non-null    object
 10  Unnamed: 10   99 non-null     object
 11  Unnamed: 11   98 non-null     object
 12  Unnamed: 12   3 non-null      object
 13  Unnamed: 13   3 non-null      object
dtypes: float64(7), object(7)
memory usage: 11.8+ KB
```

Removing the columns named, 'Unnamed (5-13)', as it is not relevant with the problem statements

	Dentist	Method	Alloy	Temp	Response
0	1.0	1.0	1.0	1500.0	813.0
1	1.0	1.0	1.0	1600.0	792.0
2	1.0	1.0	1.0	1700.0	792.0
3	1.0	1.0	2.0	1500.0	907.0
4	1.0	1.0	2.0	1600.0	792.0
...	...	...	...	...	...
85	5.0	3.0	1.0	1600.0	483.0
86	5.0	3.0	1.0	1700.0	405.0
87	5.0	3.0	2.0	1500.0	536.0
88	5.0	3.0	2.0	1600.0	405.0
89	5.0	3.0	2.0	1700.0	312.0

Removing the NaN (missing) values using the dropna() method.

Inspecting the Summary Statistics of the Dataset

	Dentist	Method	Alloy	Temp	Response
count	90.000000	90.000000	90.000000	90.000000	90.000000
mean	3.000000	2.000000	1.500000	1600.000000	741.777778
std	1.422136	0.821071	0.502801	82.107083	145.767845
min	1.000000	1.000000	1.000000	1500.000000	289.000000
25%	2.000000	1.000000	1.000000	1500.000000	698.000000
50%	3.000000	2.000000	1.500000	1600.000000	767.000000
75%	4.000000	3.000000	2.000000	1700.000000	824.000000
max	5.000000	3.000000	2.000000	1700.000000	1115.000000

Observations

Dentist: The responses range from 1 to 5, with a mean of 3. This indicates that most responses are centred around the higher end of the scale.

Method: The responses range from 1 to 3, with a mean of 2. It seems that there's a relatively even distribution among the three methods, with a slight preference for method 2.

Alloy: The responses range from 1 to 2, with a mean of 1.5. It appears that both alloys are commonly used, with no significant preference for one over the other.

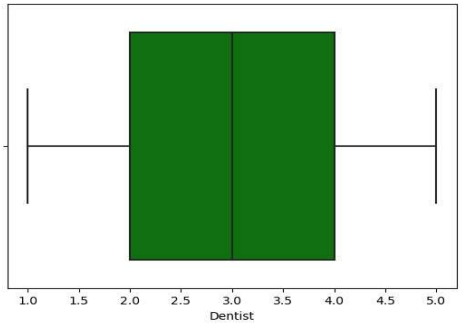
Temperature: The temperatures range from 1500 to 1700 degrees, with a mean of 1600 degrees. This suggests that the majority of responses fall within this temperature range, with 1600 degrees being the most common.

Response: The response times range from 289 to 1115 units, with a mean of approximately 741.77 units. There seems to be some variability in response times, with a notable standard deviation of 145.77 units. We'll be treating the outliers if any.

Overall, the data indicates that there's diversity in the methods and alloys used by dentists, but there's some consistency in the temperature and response time of their processes.

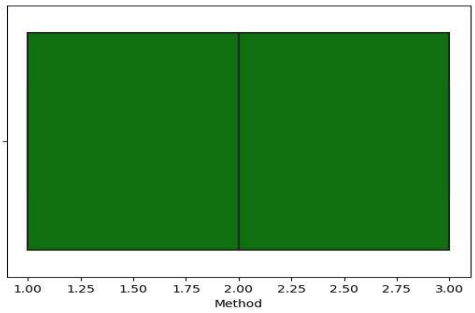
Univariate Analysis

For performing Univariate analysis, we will take a look at the Boxplots to get better understanding of the distributions.



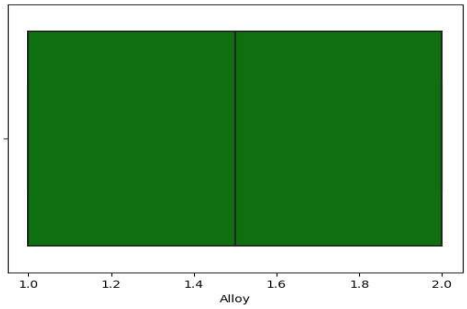
Observations on Dentist

- **Follows a normal distribution**
- **No outliers**



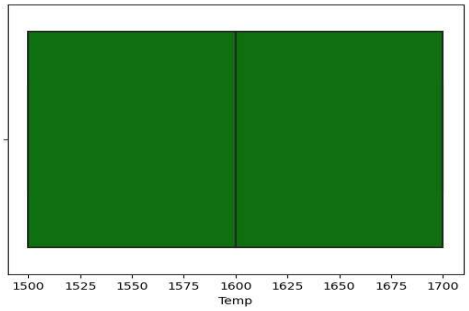
Observations on Method

- **Follows a normal distribution**
- **No Outliers**



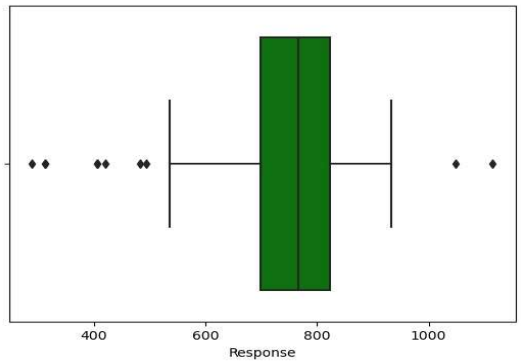
Observations on Alloy

- **Follows a normal distribution**
- **No Outliers**



Observations on Temp

- **Follows a normal distribution**
- **No Outliers**

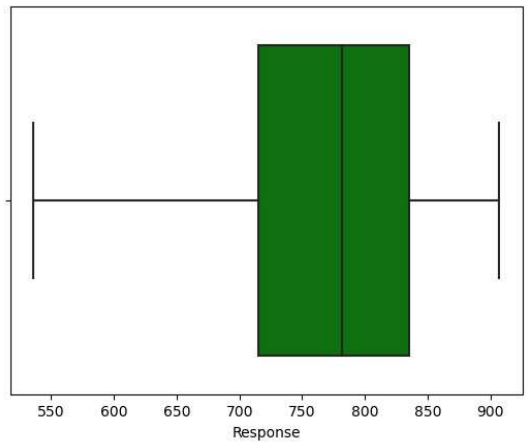


Observations on Response

- **Skewed to the right**
- **Few outliers**

Referring to the above box plots it seems that the variables 'Response' has outliers present in the variables.

These outlier values need to be treated and there are several ways of treating them:



- Drop the outlier value
- Replace the outlier value using the IQR
- We can treat the outlier's using percentile or with Inter quartile Range

Here, we are using IQR to replace outliers

5.1 How does the hardness of implants vary depending on dentists?

Hypotheses: For each alloy type, we can set up the following hypotheses:

Null Hypothesis (H0): There is no significant difference in the mean hardness of implants among dentists for the given alloy type.

Alternative Hypothesis (H1): There is a significant difference in the mean hardness of implants among dentists for the given alloy type.

ANOVA table for Alloy 1:

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	14633.1	3658.275000	0.953882	0.444837
Residual	35.0	134230.0	3835.142857	NaN	NaN

The ANOVA test examines whether there are significant differences in mean implant hardness among different dentists for Alloy 1.

The p-value (PR(>F)) associated with the Dentist factor is 0.444837, which is greater than the typical significance level of 0.05. Therefore, we fail to reject the null hypothesis.

The F-statistic is 0.953882, indicating that there is not enough evidence to suggest significant differences in mean implant hardness among dentists for Alloy 1.

In conclusion, based on this ANOVA analysis, there is no statistically significant variation in mean implant hardness among different dentists for Alloy 1.

```
ANOVA table for Alloy 2:
              df         sum_sq       mean_sq         F    PR(>F)
C(Dentist)   4.0    5014.531746   1253.632937  0.123079  0.973296
Residual    37.0  376867.873016  10185.618190       NaN       NaN
```

The ANOVA test examines whether there are significant differences in mean implant hardness among different dentists for Alloy 2.

The p-value (PR(>F)) associated with the Dentist factor is 0.973296, which is much greater than the typical significance level of 0.05. Therefore, we fail to reject the null hypothesis.

The F-statistic is 0.123079, indicating that there is not enough evidence to suggest significant differences in mean implant hardness among dentists for Alloy 2.

In conclusion, based on this ANOVA analysis, there is no statistically significant variation in mean implant hardness among different dentists for Alloy 2.

5.2 How does the hardness of implants vary depending on methods?

Null Hypothesis (H0): *There is no significant difference in the mean hardness of implants among different methods for the given alloy type.*

Alternative Hypothesis (H1)*: There is a significant difference in the mean hardness of implants among different methods for the given alloy type.*

```
ANOVA table for Alloy 1:
             df         sum_sq      mean_sq         F    PR(>F)
C(Method)   2.0    2622.966667  1311.483333  0.331816  0.719732
Residual   37.0  146240.133333  3952.436036       NaN       NaN
```

The ANOVA test examines whether there are significant differences in mean implant hardness among different methods for Alloy 1.

The p-value (PR(>F)) associated with the Method factor is 0.719732, which is much greater than the typical significance level of 0.05. Therefore, we fail to reject the null hypothesis.

The F-statistic is 0.331816, indicating that there is not enough evidence to suggest significant differences in mean implant hardness among methods for Alloy 1.

In conclusion, based on this ANOVA analysis, there is no statistically significant variation in mean implant hardness among different methods for Alloy 1.

```
ANOVA table for Alloy 2:
             df         sum_sq       mean_sq          F    PR(>F)
C(Method)   2.0  165291.221429  82645.610714  14.881394  0.000016
Residual   39.0  216591.183333   5553.620085        NaN       NaN
```

The ANOVA test examines whether there are significant differences in mean implant hardness among different methods for Alloy 2.

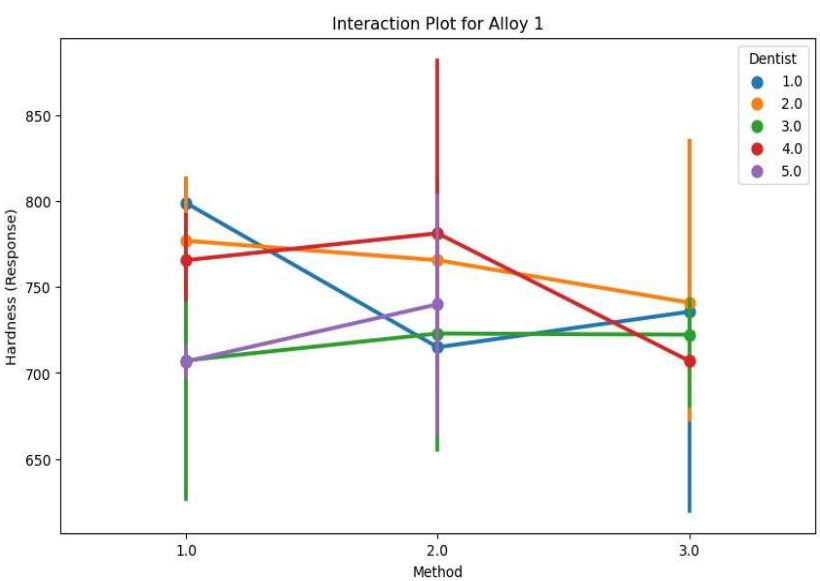
The p-value (PR(>F)) associated with the Method factor is very small (0.000016), which is much less than the typical significance level of 0.05. Therefore, we reject the null hypothesis.

The F-statistic is 14.881394, indicating that there is evidence to suggest significant differences in mean implant hardness among methods for Alloy 2.

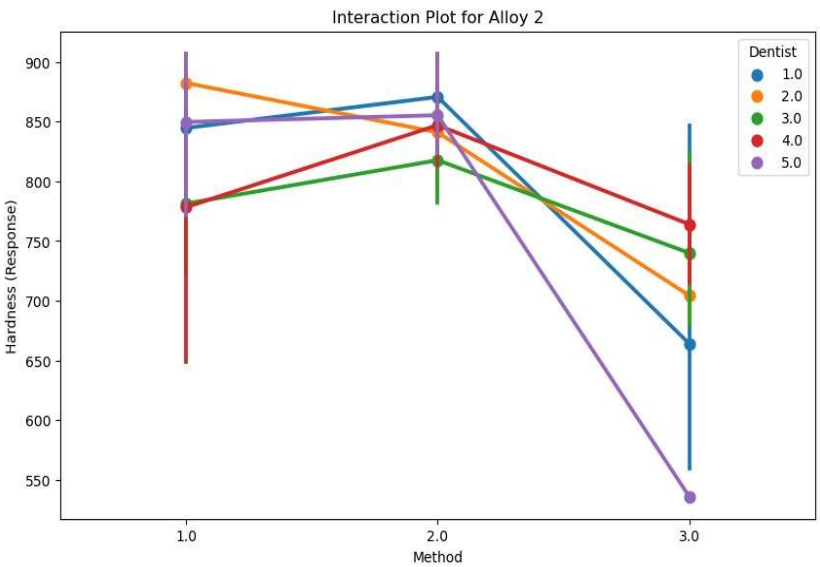
In conclusion, based on this ANOVA analysis, there is statistically significant variation in mean implant hardness among different methods for Alloy 2.

5.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

Create Interaction Plot



Conclusion: There is a significant interaction between dentist and method for Alloy 1. The method's effect on hardness depends on which dentist is performing the implant.



Conclusion: There is no significant interaction between dentist and method for Alloy 2. The method's effect on hardness is similar for all dentists.

5.4 How does the hardness of implants vary depending on dentists and methods together?

Null Hypotheses (H0):

There is no difference in hardness of implants among different dentists.

There is no difference in hardness of implants among different methods.

There is no interaction effect between dentists and methods on the hardness of implants.

Alternative Hypotheses (H1):

There is a difference in hardness of implants among different dentists.

There is a difference in hardness of implants among different methods.

There is an interaction effect between dentists and methods on the hardness of implants.

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	14633.100000	3658.275000	0.831378	0.517482
C(Method)	2.0	3390.716667	1695.358333	0.385286	0.684075
C(Dentist):C(Method)	8.0	17024.950117	2128.118765	0.483635	0.856556
Residual	26.0	114406.666667	4400.256410	NaN	NaN

Dentist

The p-value (0.517482) is greater than the significance level (usually 0.05), indicating that we fail to reject the null hypothesis.

Conclusion: There is no significant difference in the hardness of implants among different dentists.

Method

The p-value (0.684075) is greater than the significance level, indicating that we fail to reject the null hypothesis.

Conclusion: There is no significant difference in the hardness of implants among different methods.

Dentist (Interaction between Dentist and Method):

The p-value (0.856556) is greater than the significance level, indicating that we fail to reject the null hypothesis.

Conclusion: There is no significant interaction effect between dentists and methods on the hardness of implants.

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	5014.531746	1253.632937	0.236412	0.915327
C(Method)	2.0	165209.192674	82604.596337	15.577717	0.000032
C(Dentist):C(Method)	8.0	68484.680342	8560.585043	1.614370	0.167194
Residual	27.0	143174.000000	5302.740741	NaN	NaN

Dentist: The p-value (0.915327) is much greater than the significance level (usually 0.05), indicating that we fail to reject the null hypothesis.

Conclusion: There is no significant difference in the hardness of implants among different dentists.

Method: The p-value (0.000032) is much smaller than the significance level, indicating that we reject the null hypothesis***.***

Conclusion: There is a significant difference in the hardness of implants among different methods.

Dentist (Interaction between Dentist and Method): The p-value (0.167194) is greater than the significance level, indicating that we fail to reject the null hypothesis.

Conclusion: There is no significant interaction effect between dentists and methods on the hardness of implants.